# Independent BothBlocks16: mixed objects | all5 arms frozen before generation

| | GT | CIE | Dual | Global | G19 material | **BothBlocks** |
|---|---|---|---|---|---|---|
| 0001 annulus<br>high chi 18.0 | | N 0.094 \| IoU 0.801 | N 0.093 \| IoU 0.810 | N 0.093 \| IoU 0.806 | N 0.094 \| IoU 0.801 | **N 0.094 \| IoU 0.802** |
| 0003 slotted shell<br>high chi 12.0 | | N 0.413 \| IoU 0.235 | N 0.101 \| IoU 0.755 | N 0.099 \| IoU 0.771 | N 0.099 \| IoU 0.778 | **N 0.100 \| IoU 0.766** |
| 0005 multi island<br>high chi 18.0 | | N 0.076 \| IoU 0.500 | N 0.077 \| IoU 0.489 | N 0.077 \| IoU 0.500 | N 0.077 \| IoU 0.484 | **N 0.076 \| IoU 0.489** |
| 0007 curved bars<br>high chi 12.0 | | N 0.101 \| IoU 0.728 | N 0.101 \| IoU 0.712 | N 0.101 \| IoU 0.728 | N 0.100 \| IoU 0.733 | **N 0.102 \| IoU 0.733** |
| 0009 comb<br>high chi 18.0 | | N 0.118 \| IoU 0.477 | N 0.117 \| IoU 0.477 | N 0.118 \| IoU 0.485 | N 0.117 \| IoU 0.477 | **N 0.117 \| IoU 0.483** |
| 0011 coated inclusion<br>high chi 12.0 | | N 0.073 \| IoU 0.860 | N 0.073 \| IoU 0.868 | N 0.073 \| IoU 0.851 | N 0.072 \| IoU 0.851 | **N 0.073 \| IoU 0.860** |
| 0013 perforated star<br>high chi 18.0 | | N 0.113 \| IoU 0.618 | N 0.109 \| IoU 0.665 | N 0.109 \| IoU 0.643 | N 0.109 \| IoU 0.662 | **N 0.109 \| IoU 0.639** |
| 0015 bridge<br>high chi 12.0 | | N 0.068 \| IoU 0.810 | N 0.068 \| IoU 0.817 | N 0.068 \| IoU 0.810 | N 0.068 \| IoU 0.810 | **N 0.068 \| IoU 0.817** |

Real material contrast chi; common 0–20 scale

20.0 17.5 15.0 12.5 10.0 7.5 5.0 2.5 0.0

N = amplitude NRMSE. Red = material gate FAIL. All predictions selected by public Phi and sealed before GT scoring.